Assignment 2:

Using a feature from the power spectrum of the EEG to assess depth of anesthesia

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1. Introduction:

Anesthesia is a medical intervention that prevents patients from feeling pain during procedures like surgery, certain screening and diagnostic tests and dental work. Depending on the type of pain relief need, anesthesiologists can deliver anesthetics through several methods. Method can be for example gas that the patient inhales through mask, intravenous line with a needle inserted into a vein, eye drops or skin patch. Advanced anesthesia has allowed for lifesaving surgeries that would be impossible otherwise. Nowadays anesthetics are usually very safe but there are a few risks involved, including breathing problems and allergic reactions. [1]

Anesthetic depth is the degree to which the central nervous systems is depressed by a general anaesthetic agent, depending on the potency of anasthetic agent and the concentration in which is administered. An EEG based monitors would enable objective, reproducible and continuos measurement of anesthetic depth, even the patient is fully paralysed. Monitors analyse the potential fluctuations of EEG signals measures from the patient's forehead. After amplification and conversion of the analog EEG signal to the digital domain, various signal processing algorithms are applied to the frequency, amplitude, latency, and phase relationship data derived from the raw EEG. After that singe number is generated, which is scaled between 0 to 100 and called the index of depth. [2]

2. Methods

There are many concepts of analytical techniques to quantifying the irregularity of stochastics signals like EEG. One of these is entropy. Physically entropy is defined by the logarithm of number of microstates available to a thermodynamic system and describes the amount of disorder in a system. In biosignals entropy describes the irregularity, complexity or unpredictability characteristics of a signal. For example, a clear sin-signal which includes only one frequency and one amplitude, has an entropy value of zero. On the other hand, signal which includes a huge number of different frequencies and amplitudes has a bis entropy value. [3]

The method used in this exercise is called Spectral entropy. For calculating the spectral entropy, firstly, we must filter our EEG data with high pass filter to filter out low frequencies which affect the interference. After that, we have to take PSD of our signal and that is done with the Welch method in *entropy_eeg function*. When we have the PSD, we must create a function which takes PSD of one window at the time as an input. Another input is normalization factor, which corresponds to the length of the PSD. My function calculates the spectral entropy by normalizing the PSD of the window and then the entropy of the window is gotten with the following equation.

$$H = -\frac{\sum P * \log(P)}{sf}, \tag{1}$$

where P is the normalized PSD of the window and sf is the normalization factor. When the entropy of the first window is calculated, we move to the next one with the loop until whole EEG data is gone through. Both functions attached. In this case the frequency components of PSD have chosen between 0.8Hz and 32Hz because that is the frequency cap where the brain action normally is. Now the normalized value of spectral entropy between 0 and 1 tells the depth of the anesthesia. When the value is near 1, brain activity is big, and patient is awake. On the other hand, when value approaches to small numbers, brain activity is very simple. [4]

3. Results

EEG signal and before and after annotations signals presented in figure one. Before and after signals has multiplied by 100, to get better visualization and comparison to EEG signal.

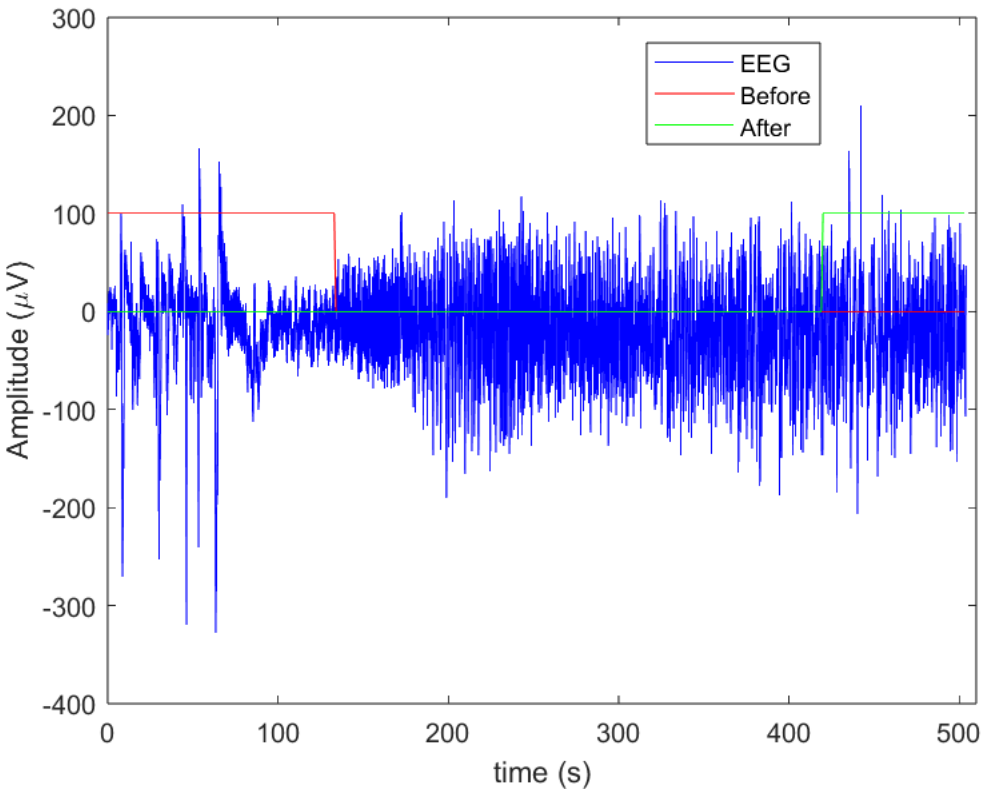

Figure 1: EEG signal with the before and after anesthesia annotations

By viewing the raw EEG signal, it is noticeable that the amplitude varies much more at the beginning and at the end of the measurement than at the mid segment. By a closer look, also frequencies vary more. Based on that, I could say that awake parts are at the beginning and at the end and the anesthesia part is between these two. That is also what before and after annotations show. The sampling frequency of the signal can be calculated when the time of the measurement and the number of samples is known with the following way.

$$t_s = \frac{t_m}{N} \tag{2}$$

$$f_s = \frac{1}{t_s}, \tag{3}$$

where t_m is measurement time, N is the number of samples and t_s is time per sample. With that method the sampling frequency is about 128Hz.

Next step is to filter our data with highpass filter with cutoff frequency 0.5Hz so too small frequencies do not disturb our signal. I chose FIR (finite impulse response) filter because these are always stable because the present output does not hold any relationship with the previous output. These can implement linear-phase filtering which mean that the filter has no phase shift across the frequency band. [3]

The phase and magnitude responses of my filter presented in figure 2.

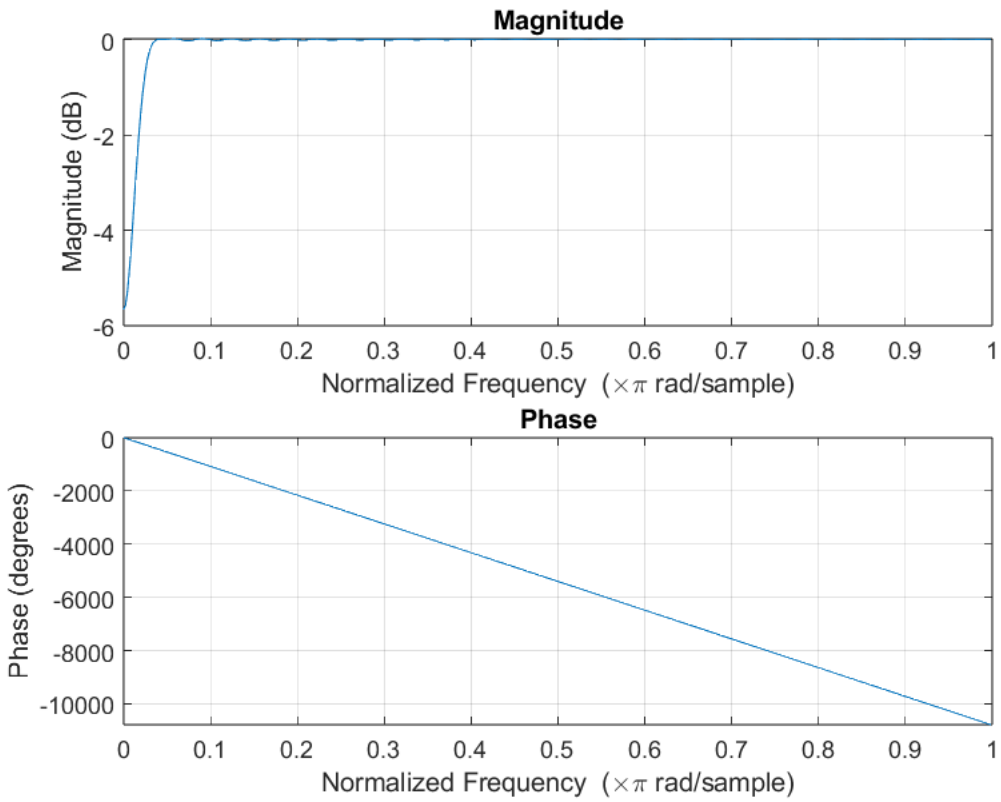

Figure 2: Magnitude and phase responses for highpass filter with cutoff frequency 0.5Hz and filter order 120.

From phase response figure, we can say that the phase shift is linear as it should be and based on magnitude figure, we can say that our filter attenuates frequencies smaller than 0.5Hz efficiently so filter works as it should work. Original and filtered EEG are plotted in figure 3.

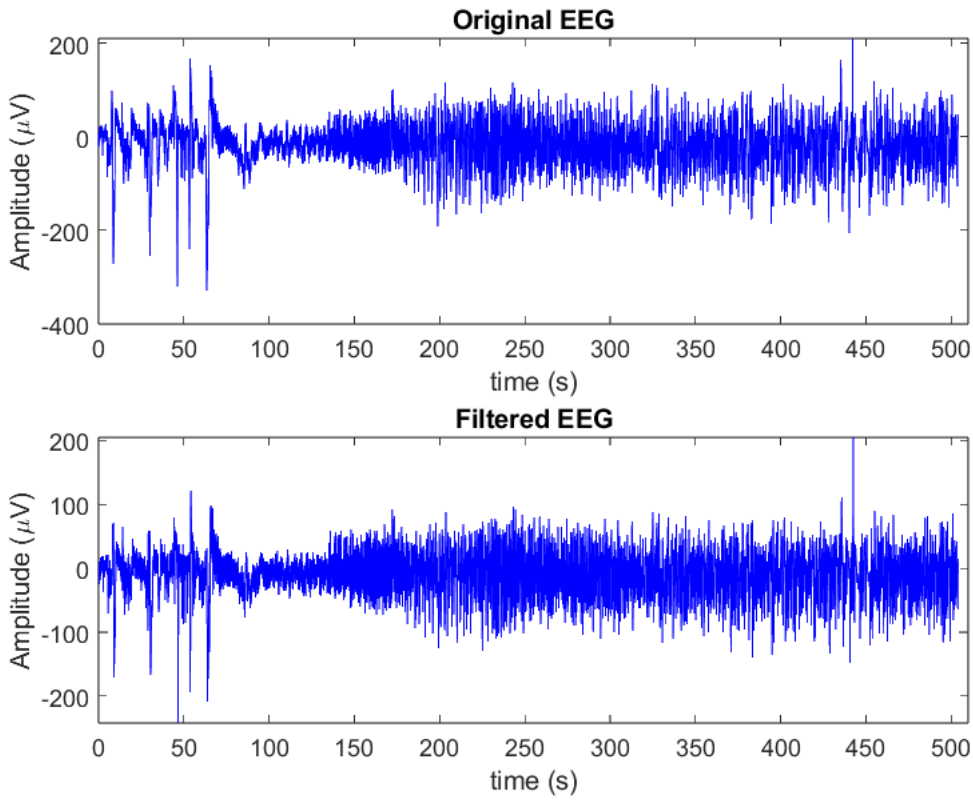

Figure 3: Original and filtered EEG

With a quick glance, signals look quite similar. However, with a closer look we can see that specifically at the awaking parts, peaks are not as high as in original signal and at the anesthesia part, higher frequencies are little bit more visible.

In figure 4, we have the spectral entropy values multiplied by 100 and BIS-monitor values with the before and after annotations.

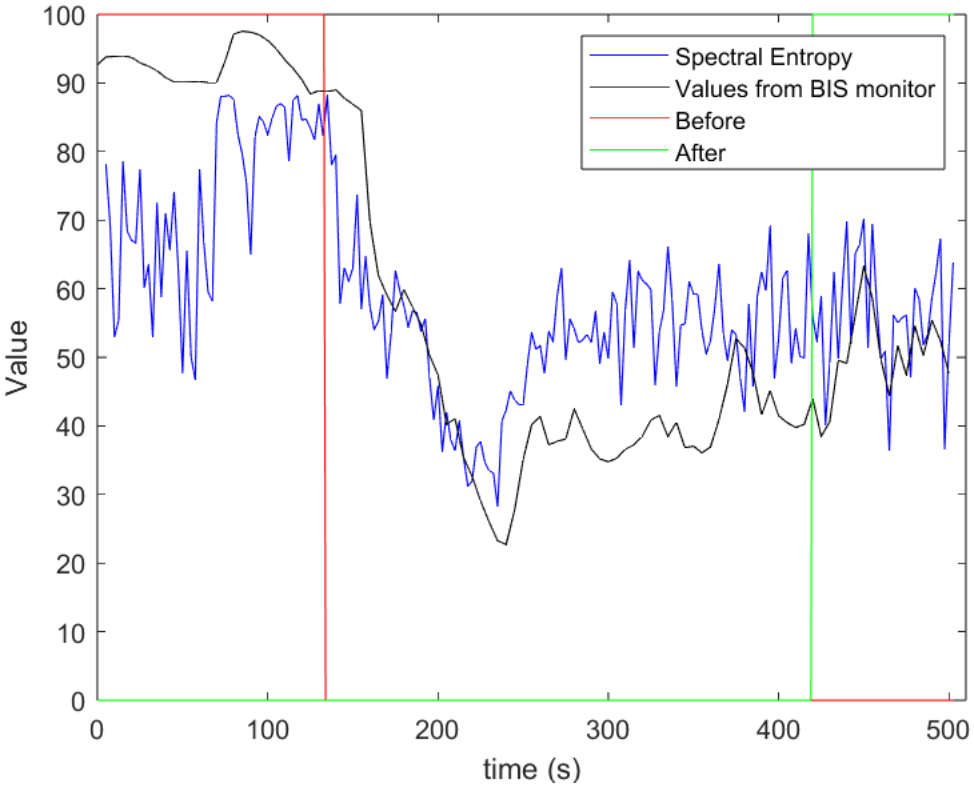

Figure 4: BIS monitor values and Spectral Entropy

Spectral Entropy values are multiplied by 100 so these are comparable with BIS values. From picture, we can see that values are higher before anesthesia, but decrease quickly when anesthesia has started. It is noticeable that after anesthesia, values no dot return back to level before anesthesia during measurement time. So, effects of anesthesia are seen in brain activity still after awakening. Trend between BIS and spectral entropy looks similar, but Spectral Entropy signal is much more detailed than BIS values which might be due to different number of datapoints in averaging. However, before anesthesia, BIS values are a little bit higher than entropy values while in anesthesia part, entropy values are higher.

BIS and specral entropy are not completely same because methods are based on different mathematical and physical principles. BIS uses the compex algorithm which is based on time and frequency domain analysis. Spectral entropy creates a frequency spectrum and generates the number which describes the irregularity and complexity of signal.

Also, one reason for the difference is the electromyographic (EMG) component. While BIS monitor measures values from patient's forehead, it includes significant EMG component which is affected by muscle activity. The EMG signal has a wide noise-like spectrum in range of frequencies from 32Hz to 47Hz. State entropy values are resistant to sudden muscle activity because it notices only the frequency distribution of EEG. State Entropy also focuses more to certain frequency ranges and that is why it is not as comprehensive as BIS. [2]

As the entropy signal look good, there are still ways to improve it. In pre-processing, we could for example use more effective filtering methods and when calculating the spectral entropy, different methods or parameters could be used. In post-processing we could smooth our signal, so it does not look as detailed as it looks now but trend and bigger changes would still be visible.

4. Conclusion

Anesthesia plays a crucial role in medicine nowadays, ensuring patients remain pain-free during surgeries and various properties. Estimating the depth of anesthesia is crucial for patient's safety so it is very important to have trustable tools like BIS and spectral entropy for it. I personally gained a huge amount of information about anesthesia during this exercise. Additionally, I learned, how to estimate the depth of it with various methods and differences between those. My Matlab-skills also improved a lot while processing the data.

5. Resources

[1] National Institute of General Medical Sciences, “Anesthesia” (2024), [Referred 5.4.2024]. https://www.nigms.nih.gov/education/fact-sheets/Pages/anesthesia.aspx

[2] DD Rani, SS Harsoor, National Library of Medicine, “Depth of general anaesthesia monitors” (2012). [Referred 5.4.2024] https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3530997/

[3] MiniDSP, “FIR vs IIR filtering”, [Referred 5.4.2024]. https://www.minidsp.com/applications/dsp-basics/fir-vs-iir-filtering

[4] Acta Anaesthesiologica Scandinavica, ”Description of the Entropy algorithm as applied in Datex-Ohmeda S/5 Entropy Module”. [Referred 8.4.2024]. SpectralEntropyDOA.pdf

6. Matlab codes

```
function H = spectral_entropy(PSD, sf)
    % Inputs:
    % PSD - Power Spectral Density of window
    % sd - scaling factor, needed for normalization of spectral entropy
    %
    %calculates spectral entropy
    %
    % Output:
    %  H - Spectral entropy

    %normalization
    P=PSD./sum(PSD);

    %calculate the entropy
    H = -sum(P.*log(P))/sf;

end
```

```
load("case_1.mat");
```

1)

```
figure
plot(eeg_time,eeg_ch,"b-")
hold on
plot(annot_time,100*before,"r-")
hold on
plot(annot_time,100*after,"g-")
legend("EEG","Before","After")
xlabel("time (s)")
ylabel("Amplitude (\muV)")
xlim([0 510])
hold off
%anesthesia period between 133s-420s
```

2)

```
measure_time = eeg_time(64512);
samples = 64512;
sample_time = measure_time/samples;
fs_eeg = round(1/sample_time,0)
```

Sampling frequency is 128Hz

3)

```
%FIR highpass filter with cutoff frequency 0.5Hz
N = 120
fc = 0.5/(fs_eeg/2) %normalized

b = fir1(N, fc, "high");
figure
freqz(b)
eeg_f = filter(b,1,eeg_ch);
```

4)

```
figure
subplot(2,1,1)
plot(eeg_time,eeg_ch,"b-")
title("Original EEG")
ylabel("Amplitude (\muV)")
xlabel("time (s)")
xlim([0 510])
subplot(2,1,2)
plot(eeg_time,eeg_f,"b-")'
title("Filtered EEG")
ylabel("Amplitude (\muV)")
xlabel("time (s)")
xlim([0 510])

%Affect to the awaking parts, peaks are not as high as an original signal
```

6)

```
[H,H_t] = entropy_eeg(eeg_f,fs_eeg);
```

7)

```
figure
plot(H_t,100*H,"b-")
hold on
plot(bis_time,bis_ch,"k-")
hold on
plot(annot_time,100*before,"r-")
hold on
plot(annot_time,100*after,"g-")
hold off
xlabel("time (s)")
ylabel("Value")
legend("Spectral Entropy", "Values from BIS monitor","Before","After")
xlim([0 510])
```

```
function [H,t] = entropy_eeg(eeg,fs)
%   Computes spectral entropy for EEG


% inputs :
% eeg - EEG signal
% fs    - sampling frequency (Hz)

% outputs:
% H - spectral entropy over 0.5-32 Hz
% t - time axis (seconds)

eeg_length = length(eeg); % number of samples in (filtered) EEG
wlen = 5; % window length to calculate PSD over (seconds)
overlap = 50; % percentage overlap for windows
H_length = round(1 + eeg_length/(wlen*fs*((100-overlap)/100))); % estimate the number of power spectrum estimations
H = zeros(H_length, 1)*NaN;   % initialization array containing spectral entropy values
t = zeros(H_length, 1)*NaN; % initialization array containing time instances at which spectral entropy values are calculated

% spectrum calculation over first window
t0 = 1;  % first sample of the window
t1 = wlen * fs; % last sample
step = round(fs*wlen*(100-overlap)/100); % number of samples to shift the window each step
cnt = 1; % step counter
% [PSD,f] = psd(eeg(t0:t1),t1-t0+1,fs,hanning(t1-t0+1),0); %first PSD calculation (alternatively, use spectrum.welch)
% PSD now contains PSD values at the frequencies given in array f
[PSD, f] = pwelch(eeg(t0:t1), hanning(t1-t0+1), 0, t1-t0+1, fs);

f_low = 0.8;   % lower limit of frequency range of interest
f_high = 32.0; % upper limit of frequency range of interest
ind_f = find(f >= f_low & f <= f_high);  % find indexes of relevant frequencies in f array
sf = log(length(ind_f)); % scaling factor, needed for normalization of spectral entropy

% now move the window through the eeg data and calculate PSD and Entropy at
% each position and write to H array
```

```
while t1 < eeg_length  % as long as the last sample of the window is not beyond the EEG signal end
 % PSD = psd(eeg(t0:t1),t1-t0+1,fs,hanning(t1-t0+1),0); %calculate the PSD over a window
 % PSD has been removed

 PSD = pwelch(eeg(t0:t1), hanning(t1-t0+1), 0, t1-t0+1, fs);
 t(cnt) = t1/fs; % time instance of PSD (and entropy calculation), defined as end of window
 H(cnt) = spectral_entropy(PSD(ind_f),sf);  % calculate normalized spectral entropy over ind_f freq band - WRITE FUNCTION CONTENTS YOURSELF
 t0 = t0 + step; % update start position of window
 t1 = t1 + step; % update end position of window
 cnt = cnt + 1; % update step counter

end
```

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